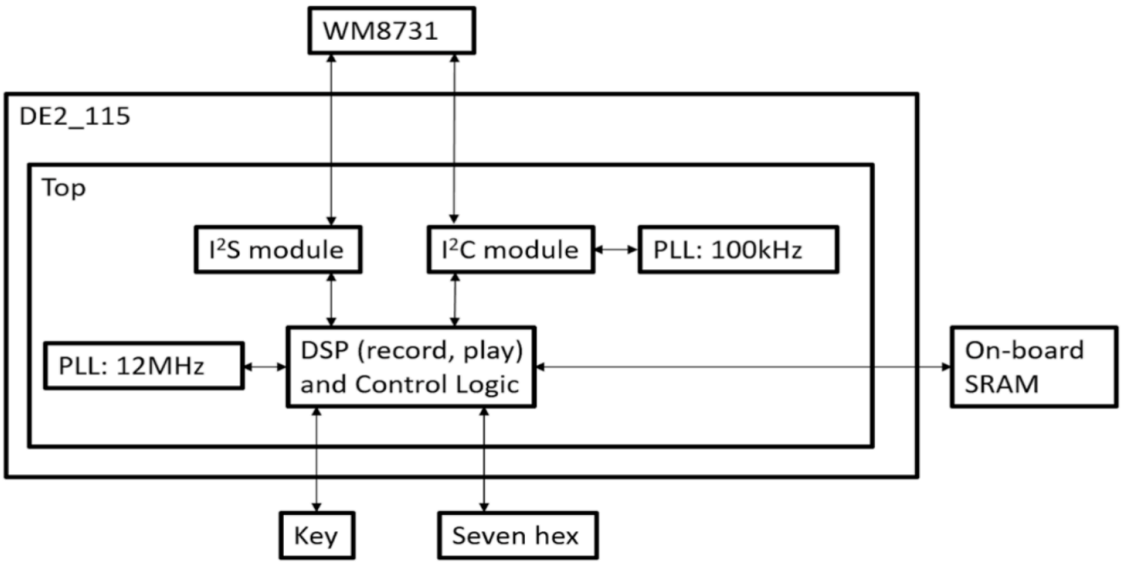
DCLab Lab3 Report Group12

File Structure

```
team12_lab3
|- team12_lab3_report.pdf
|- Altpll.qsys
|- Altpll
  |- synthesis
    |- Altpll.v
    |- Altpll.qip
|- src
  |- DE2_115
    |- DE2_115.qsf
    |- DE2_115.sdc
    |- DE2_115.sv
    |- Debounce.sv
    |- SevenHexDecoder.sv
  |- Top.sv
  |- I2cInitializer.sv
  |- AudRecorder.sv
  |- AudPlayer.sv
  |- AudDsp.sv
```

System Architecture



User Manual

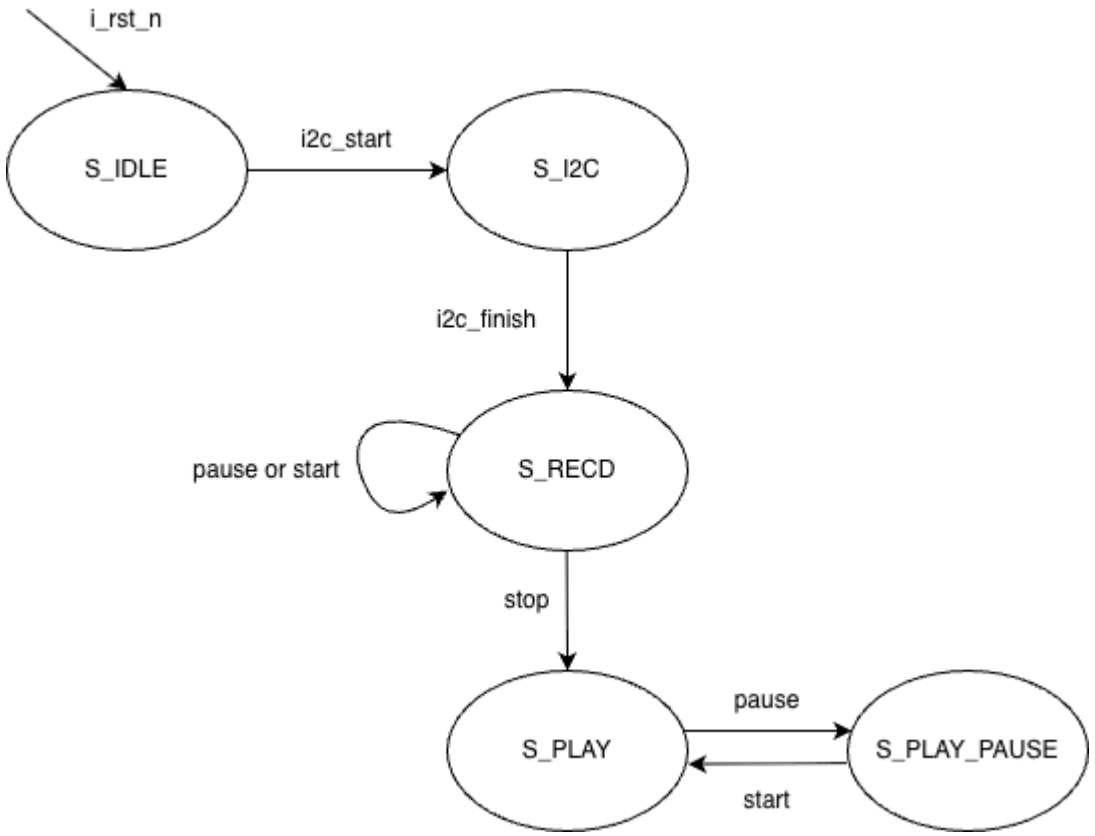
Our design uses four buttons and five switches as the user interface, three decoders and some LEDs for display.

Key0: start signal.
Key1: pause signal.
Key2: stop signal.
Key3: reset signal.
SW[2:0]: speed 1 to 8.
SW[3]: slow mode or fast mode.
SW[4]: 0 interpolation or 1 interpolation.

SevenDecoder0: record time.
SevenDecoder1: play time.
SevenDecoder2: speed.

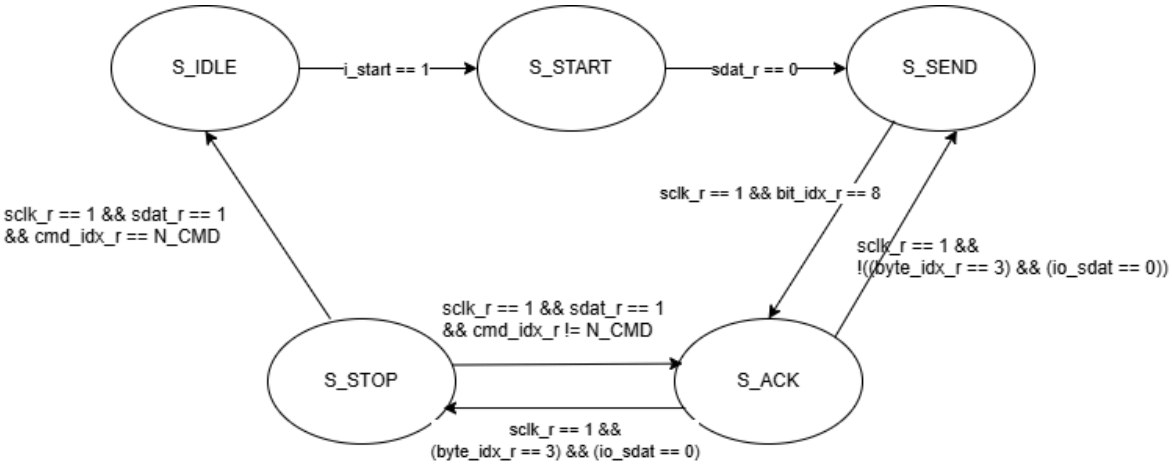
Hardware Scheduling (FSM)

1. Top



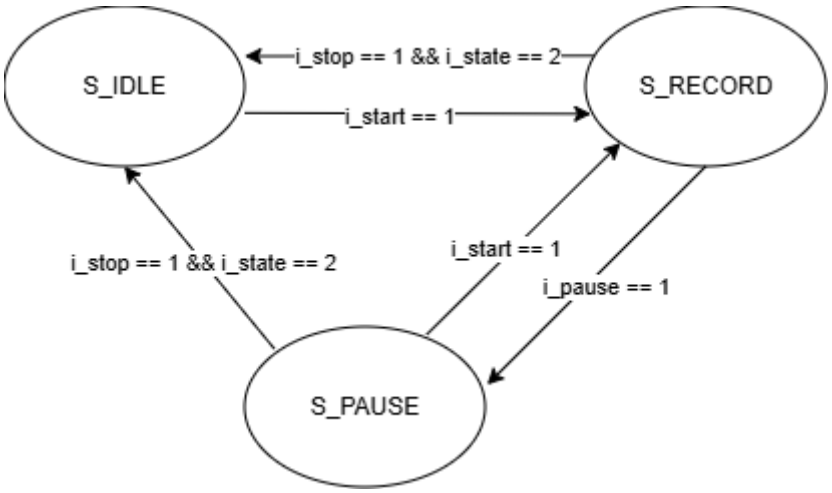
S_IDLE : Wait for start signal to begin I2C initialization
S_I2C : Wait until I2C is finished and begin recording
S_RECD : Wait until record stop and begin playing
S_PLAY : controlled by AudDSP through play_en and played by AudPlayer
S_PLAY_PAUSE : Set play_en to 0

2. I2cInitializer



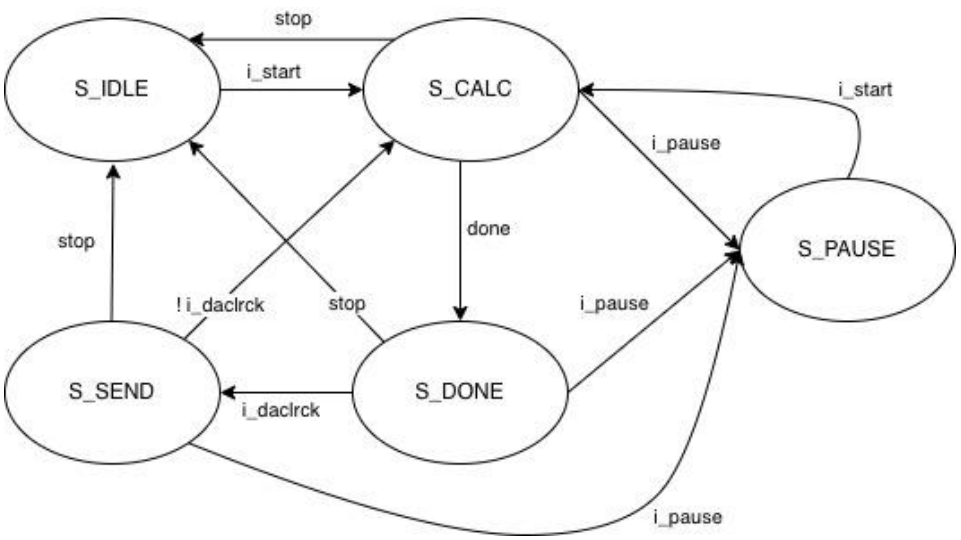
S_IDLE: Wait for start signal
S_START: Set sdat, bit index, and byte index to 0
S_SEND: Send the corresponding bit according to bit index and byte index
S_ACK: Receive ACK
S_STOP: Pull up SDAT and update cmd_idx

3. AudRecorder



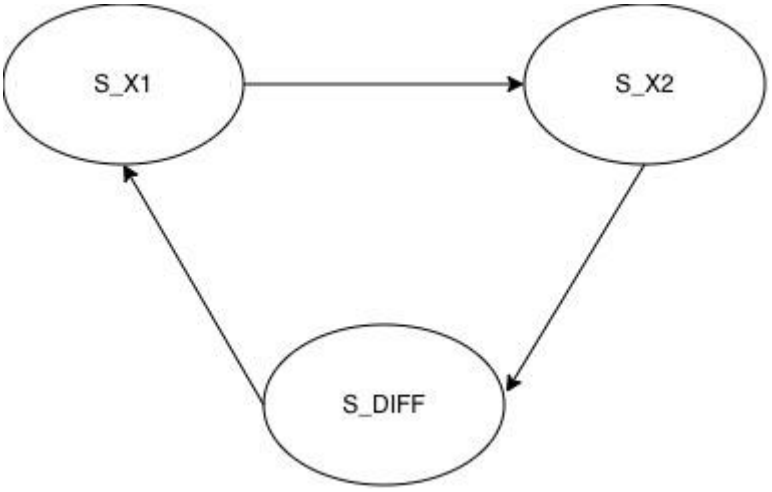
S_IDLE: Wait for start signal
S_RECORD: Store the recorded signal and the stop address
S_PAUSE: Hold the current data address and wait for start signal

4. AudDsp



AudDsp FSM

S_IDLE: Wait for start signal
S_CALC: Calculate output result; For fast mode, output sampled data at addr+i_speed; For slow-0 mode, the SRAM address is changed every i_speed cycle; For slow-1 mode, the previous output data is added by diff
S_DONE: Wait for DAC clock rise
S_SEND: Start calculating next result after DAC clock falls
S_PAUSE: Hold all register value



Slow1 substate FSM

While fast and slow-0 mode only need one cycle, slow-1 mode requires three cycles to perform interpolation:
S_X1: Fetch x1
S_X2: Fetch x2
S_DIFF: Calculate (x2-x1)/i_speed

Fitter Summary截圖

Fitter Summary	
Fitter Status	Successful - Thu Oct 30 19:07:49 2025
Quartus II 64-Bit Version	15.0.0 Build 145 04/22/2015 SJ Full Version
Revision Name	DE2_115
Top-level Entity Name	DE2_115
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	3,048 / 114,480 (3 %)
Total combinational functions	1,642 / 114,480 (1 %)
Dedicated logic registers	2,272 / 114,480 (2 %)
Total registers	2273
Total pins	518 / 529 (98 %)
Total virtual pins	0
Total memory bits	70,656 / 3,981,312 (2 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	1 / 4 (25 %)

Timing Analyzer截圖

E2_115

Timing Closure Recommendations

Summary [hide details]

This design contains failing setup paths with a worst-case slack of -6.368 ns. Run Report Timing Closure Recommendations for recommendations on how to close setup timing. For recommendations for any particular path, click the appropriate link in the table below.

Top Failing Paths [hide details]

	Slack	From	To	Recommendations
1	-6.368	Top:top0|AudRecor...top_address_r[2]	sld_signaltap:aut...gger_in_reg[132]	Report recommendations for this path
2	-6.368	Top:top0|AudRecor...top_address_r[2]	sld_signaltap:aut...data_in_reg[132]	Report recommendations for this path
3	-6.355	Top:top0|AudRecor...top_address_r[0]	sld_signaltap:aut...gger_in_reg[132]	Report recommendations for this path
4	-6.355	Top:top0|AudRecor...top_address_r[0]	sld_signaltap:aut...data_in_reg[132]	Report recommendations for this path
5	-6.350	Top:top0|AudDSP:dsp0|addr[0]	sld_signaltap:aut...gger_in_reg[132]	Report recommendations for this path

Slow 1200mV 85C Model Setup Summary

	Clock	Slack	End Point TNS
1	pll0|altpll_0|sd1|pll7|clk[0]	-6.368	-242.054
2	AUD_BCLK	-3.766	-256.554
3	altera_reserved_tck	43.791	0.000
4	pll0|altpll_0|sd1|pll7|clk[1]	76.236	0.000

Slow 1200mV 85C Model Hold Summary

	Clock	Slack	End Point TNS
1	pll0|altpll_0|sd1|pll7|clk[0]	-0.357	-0.714
2	pll0|altpll_0|sd1|pll7|clk[1]	0.402	0.000
3	altera_reserved_tck	0.403	0.000
4	AUD_BCLK	0.440	0.000

Slow 1200mV 85C Model Setup: 'pll0|altpll_0|sd1|pll7|clk[0]'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship
60	-5.722	Top:top0|AudDSP:dsp0|addr[9]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[89]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
61	-5.708	Top:top0|AudRecorder:recorder0|stop_address_r[11]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
62	-5.708	Top:top0|AudRecorder:recorder0|stop_address_r[11]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
63	-5.701	Top:top0|AudDSP:dsp0|addr[4]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[84]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
64	-5.697	Top:top0|AudDSP:dsp0|addr[4]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[84]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
65	-5.687	Top:top0|AudRecorder:recorder0|addr_counter_r[8]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[88]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
66	-5.683	Top:top0|AudRecorder:recorder0|addr_counter_r[8]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[88]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
67	-5.666	Top:top0|AudDSP:dsp0|addr[17]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[78]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
68	-5.665	Top:top0|AudDSP:dsp0|addr[17]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[78]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
69	-5.664	Top:top0|AudRecorder:recorder0|stop_address_r[14]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
70	-5.664	Top:top0|AudRecorder:recorder0|stop_address_r[14]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
71	-5.656	Top:top0|AudRecorder:recorder0|addr_counter_r[17]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[78]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
72	-5.655	Top:top0|AudRecorder:recorder0|addr_counter_r[17]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[78]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
73	-5.653	Top:top0|AudRecorder:recorder0|stop_address_r[12]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
74	-5.653	Top:top0|AudRecorder:recorder0|stop_address_r[12]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
75	-5.638	Top:top0|AudDSP:dsp0|addr[16]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
76	-5.638	Top:top0|AudDSP:dsp0|addr[16]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
77	-5.611	Top:top0|AudRecorder:recorder0|stop_address_r[9]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
78	-5.611	Top:top0|AudRecorder:recorder0|stop_address_r[9]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
79	-5.611	Top:top0|AudRecorder:recorder0|addr_counter_r[9]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[89]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
80	-5.611	Top:top0|AudRecorder:recorder0|addr_counter_r[9]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[89]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
81	-5.604	Top:top0|AudDSP:dsp0|addr[0]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[70]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
82	-5.580	Top:top0|AudDSP:dsp0|addr[2]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[82]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
83	-5.571	Top:top0|AudRecorder:recorder0|stop_address_r[17]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
84	-5.571	Top:top0|AudRecorder:recorder0|stop_address_r[17]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
85	-5.568	Top:top0|AudDSP:dsp0|addr[2]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[82]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
86	-5.543	Top:top0|AudDSP:dsp0|addr[13]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[74]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
87	-5.542	Top:top0|AudDSP:dsp0|addr[13]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[74]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
88	-5.527	Top:top0|AudRecorder:recorder0|addr_counter_r[13]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[74]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
89	-5.526	Top:top0|AudRecorder:recorder0|addr_counter_r[13]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[74]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
90	-5.512	Top:top0|AudDSP:dsp0|addr[18]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
91	-5.512	Top:top0|AudDSP:dsp0|addr[18]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
92	-5.491	Top:top0|AudDSP:dsp0|addr[17]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
93	-5.491	Top:top0|AudDSP:dsp0|addr[17]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
94	-5.484	Top:top0|AudDSP:dsp0|addr[11]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[72]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
95	-5.483	Top:top0|AudDSP:dsp0|addr[11]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[72]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
96	-5.471	Top:top0|AudDSP:dsp0|addr[10]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[71]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
97	-5.471	Top:top0|AudDSP:dsp0|addr[10]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[71]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
98	-5.458	Top:top0|AudRecorder:recorder0|addr_counter_r[11]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[72]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
99	-5.457	Top:top0|AudRecorder:recorder0|addr_counter_r[11]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[72]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001
100	-5.436	Top:top0|AudRecorder:recorder0|stop_address_r[13]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[132]	AUD_BCLK	pll0|altpll_0|sd1|pll7|clk[0]	0.001

Slow 1200mV 85C Model Setup: 'AUD_BCLK'

	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	-3.766	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[15]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.118
2	-3.766	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[4]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.118
3	-3.766	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[3]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.118
4	-3.755	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[15]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.107
5	-3.755	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[4]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.107
6	-3.755	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[3]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	6.107
7	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[14]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
8	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[13]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
9	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[12]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
10	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[11]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
11	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[10]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
12	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[9]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
13	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[8]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
14	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[7]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
15	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[6]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
16	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[5]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
17	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[2]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
18	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[1]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
19	-3.723	Top:top0|state_r[1]	Top:top0|AudPlayer:player0|data_r[0]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.987
20	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[14]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
21	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[13]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
22	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[12]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
23	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[11]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
24	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[10]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
25	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[9]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
26	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[8]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
27	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[7]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
28	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[6]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
29	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[5]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
30	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[2]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
31	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[1]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
32	-3.717	Top:top0|state_r[0]	Top:top0|AudPlayer:player0|data_r[0]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.981
33	-3.516	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[15]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	5.868
34	-3.516	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[4]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	5.868
35	-3.516	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[3]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.443	5.868
36	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[14]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
37	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[13]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
38	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[12]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
39	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[11]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
40	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[10]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
41	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[9]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709
42	-3.445	Top:top0|state_r[2]	Top:top0|AudPlayer:player0|data_r[8]	pll0|altpll_0|sd1|pll7|clk[0]	AUD_BCLK	0.001	2.355	5.709

Slow 1200mV 85C Model Hold: 'pll0|altpll_0|sd1|pll7|clk[0]'

	Slack	From Node	To Node
1	-0.357	pll0|altpll_0|sd1|pll7|clk[1]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[25]
2	-0.357	pll0|altpll_0|sd1|pll7|clk[1]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[25]
3	-0.321	pll0|altpll_0|sd1|pll7|clk[1]	sld_signaltap:auto_signaltap_0|acq_trigger_in_reg[25]
4	-0.321	pll0|altpll_0|sd1|pll7|clk[1]	sld_signaltap:auto_signaltap_0|acq_data_in_reg[25]
5	0.288	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][80]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
6	0.322	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][117]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
7	0.323	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][135]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
8	0.328	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][106]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
9	0.328	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][50]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
10	0.330	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][94]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
11	0.331	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][133]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
12	0.332	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][92]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
13	0.333	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][136]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
14	0.333	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][121]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
15	0.334	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][110]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
16	0.334	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][42]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
17	0.335	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][36]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
18	0.336	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][130]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
19	0.336	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][108]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
20	0.336	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][67]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
21	0.336	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][26]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
22	0.337	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][137]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
23	0.337	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][29]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
24	0.338	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][101]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
25	0.338	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][71]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
26	0.340	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][65]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
27	0.340	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][19]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
28	0.340	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][25]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
29	0.341	sld_signaltap:auto_signaltap_0|sld_signaltap_impl...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][1]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...24:auto_generated|ram_b
30	0.342	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][97]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
31	0.342	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][51]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
32	0.343	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][64]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
33	0.343	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][31]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
34	0.344	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][105]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
35	0.344	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][74]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
36	0.344	sld_signaltap:auto_signaltap_0|sld_signaltap_impl...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][3]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...24:auto_generated|ram_b
37	0.350	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][124]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
38	0.351	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][107]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
39	0.352	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][89]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
40	0.355	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...lb:sld_signaltap_body|acq_data_in_pipe_reg[3][134]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...:auto_generated|ram_blo
41	0.355	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][66]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl
42	0.355	sld_signaltap:auto_signaltap_0|sld_signaltap_impl:...plb:sld_signaltap_body|acq_data_in_pipe_reg[3][37]	sld_signaltap:auto_signaltap_0|sld_signaltap_imp...4:auto_generated|ram_bl

遇到的問題與解決辦法、心得與建議

1. **Slow-1 issue**
   The initial DSP design had large noise in slow-1 mode. We suspect that these large noises were caused by overflow, resulting in sharp changes and high frequency noise. We resolved the problem by adding $signed() to each calculation step.

2. **FPGA hardware limit**
   Initially, our DSP design, though synthesized successfully, failed at place and route. We found out that the problem lies in the hardware limit of the FPGA, where no more than 5 non-global control signals can be used to control another signal. The problem was solved through redesigning the data flow of the interpolation process by adding additional substates so less control signals were required.

3. **Stop address**
   At first, we didn't consider the issue that the recorder should remember the final address of recording. So the player and DSP will not stop sending data even if the address has already exceeded the correct final address. In some situations this can cause serious issues because the data in SRAM is not cleared, so it may keep playing the voice that was recorded in previous times. The solution is simply record the final address of recording, and send it to the top module. If the playing address reaches the stop address, disable the player.

4. **Stop Signal**
   In the beginning, we want the stop button to function as a renewed recording or playing in recording state and playing state respectively. However, we find that the stop button functioned during the playing state will clear the address sent during the recording state, causing the next playing process to be wrong. We then switch the stop input for recorder to recd_stop; //assign recd_stop = (state_r == S_RECD) && stop. Nonetheless it leads to an increasing noise during playing. Therefore we add another input "i_state" to AudRecorder to infer the current state and prevent clearing recording stop address in play state.